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Deliverable 2

Global Earthquake Activity

Data Analysis and Visualization Lab

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1. Introduction

Earthquakes are among the most impactful natural hazards, capable of altering landscapes, influencing human settlements, and shaping global infrastructure resilience. Understanding their patterns, frequencies, and characteristics allows scientists and analysts to evaluate seismic risks and improve preparedness strategies. Through advanced visualization and data analysis, seismic datasets reveal meaningful geographical, temporal, and geological patterns that deepen our understanding of Earth's tectonic processes.

This study focuses on analyzing global earthquake activity using multiple visual techniques to uncover relationships involving magnitude, depth, frequency, energy release, and spatial distribution across different tectonic regions.

2. Dataset Description

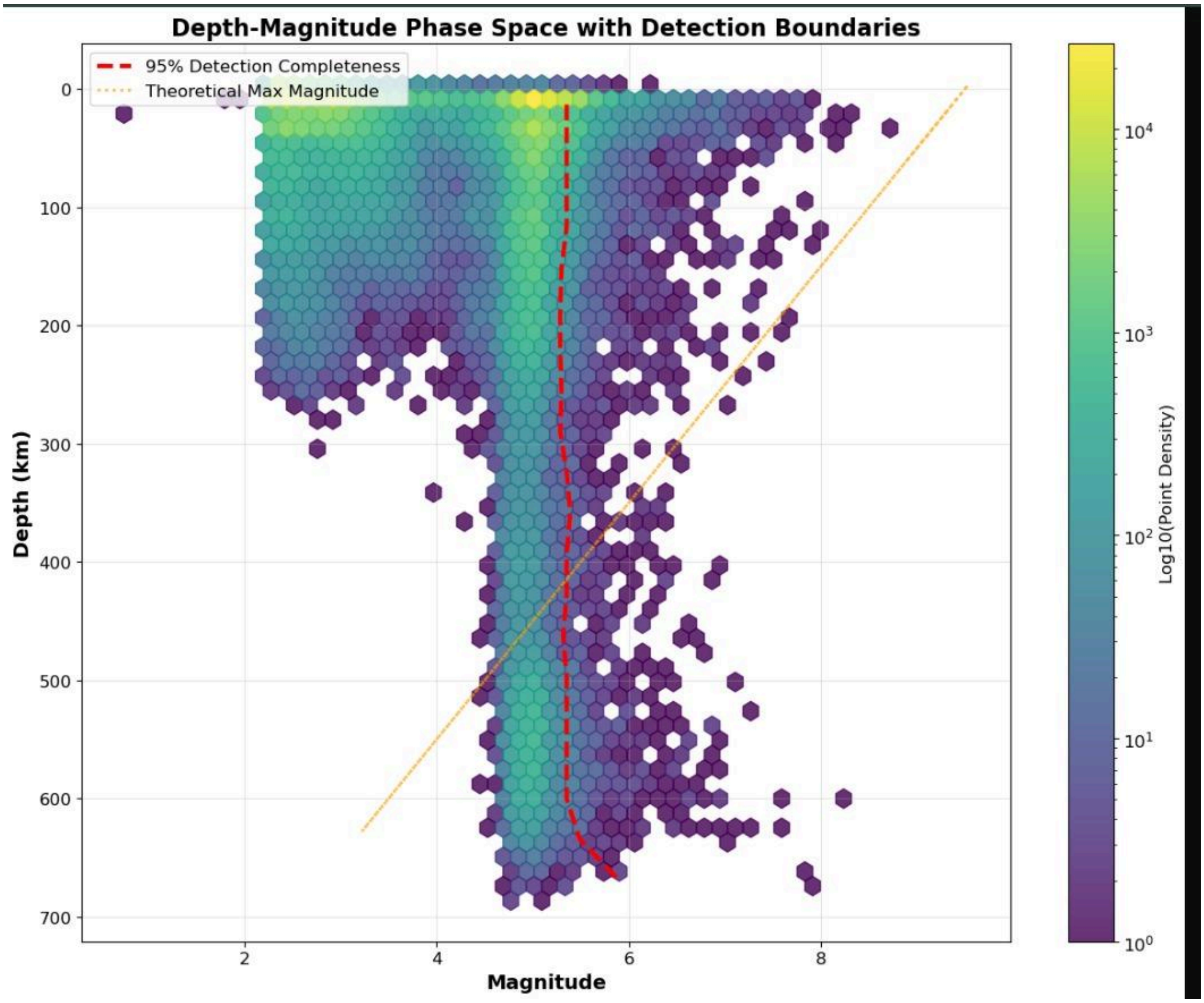
2.1. Name of Dataset: USGS Magnitude 2.5+ Earthquakes
2.2. Source: United States Geological Survey (USGS)
2.3. Dataset Link:
https://earthquake.usgs.gov/earthquakes/map/?extent=-73.82482,-354.72656&extent=84.9901,207.77344

The dataset contains global earthquake occurrences exceeding magnitude 2.5, featuring attributes such as magnitude, longitude, latitude, depth, time of occurrence, and azimuthal gap measurements. It represents a diverse and detailed seismic catalog suitable for large-scale geospatial and statistical analysis.

3. Analysis & Visualizations

3.1 Analysing Depth Magnitude with Space Detection

3.1.1. Visualisation:



3.1.2. Description:

The 95% detection completeness line represents the minimum earthquake magnitude your seismic network is capable of detecting at each depth—with 95% confidence. In other words: The red line shows the smallest magnitude that your dataset reliably includes. Anything below the line is not consistently detected and is likely missing from the data.

How to Read the Detection Line

➤ Look at a specific depth (y-axis)

Move horizontally from that depth until you hit the red dashed line. Where it intersects the x-axis (magnitude) tells you: “At this depth, earthquakes smaller than this magnitude are probably NOT detected.”

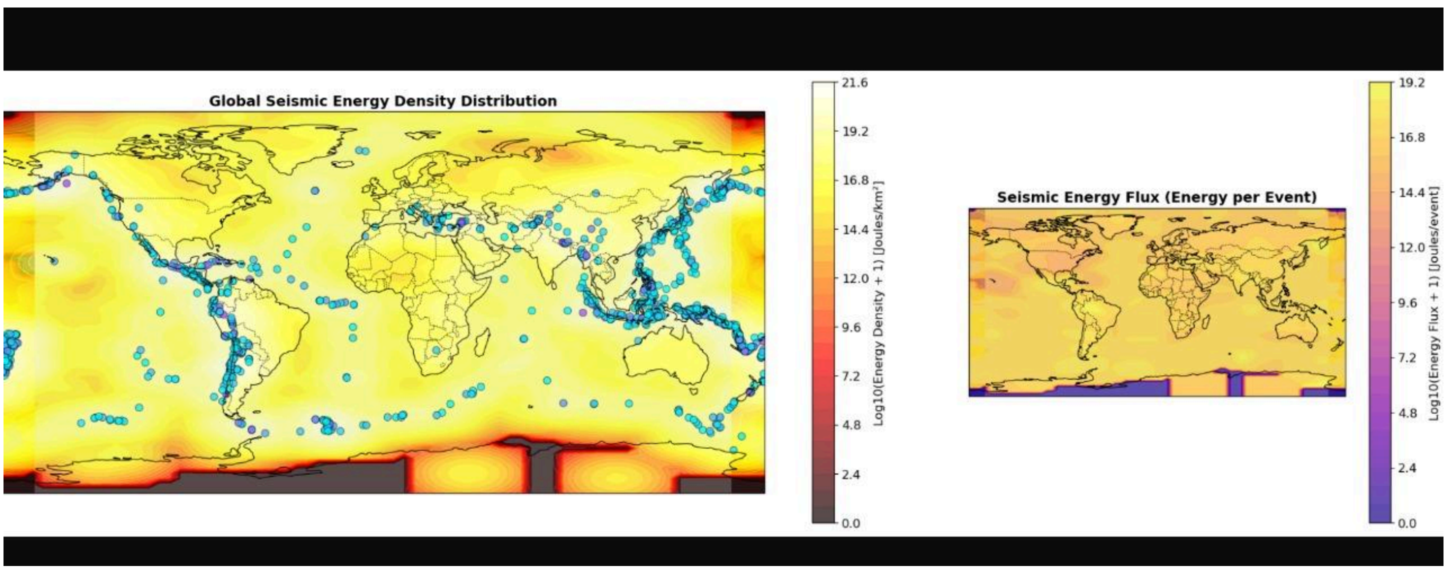
Examples: If at 100 km depth the red line sits around M ~4.0 → Your catalog does not reliably detect quakes smaller than 4 at that depth.
If at 400 km depth the red line increases to M ~5.2 → Only larger quakes are detected deep inside the Earth.

Which Parts of the Dataset Are Affected by Detection Limits?
➤ All earthquake points that fall below the red dashed line

Shallow depths (<50 km): The red line is low → You detect many small earthquakes (high completeness)
Intermediate depths (100–300 km): The line rises → You lose most small events
Deep earthquakes (>300 km): The red line is high → Only moderate or large events are recorded

3.2 Global Seismic Energy Distribution

3.2.1. Figure



3.2.2. Description:

The figure presents two global maps that illustrate how earthquake energy rather than earthquake counts is distributed across the planet, emphasizing that tectonic behavior is fundamentally driven by energy release. The first map shows global seismic energy density, highlighting where the most total energy has been released per square kilometer. Bright yellow and white areas mark extremely high energy release, red and orange indicate high energy, and dark regions represent low or negligible energy, with blue dots marking epicenters.

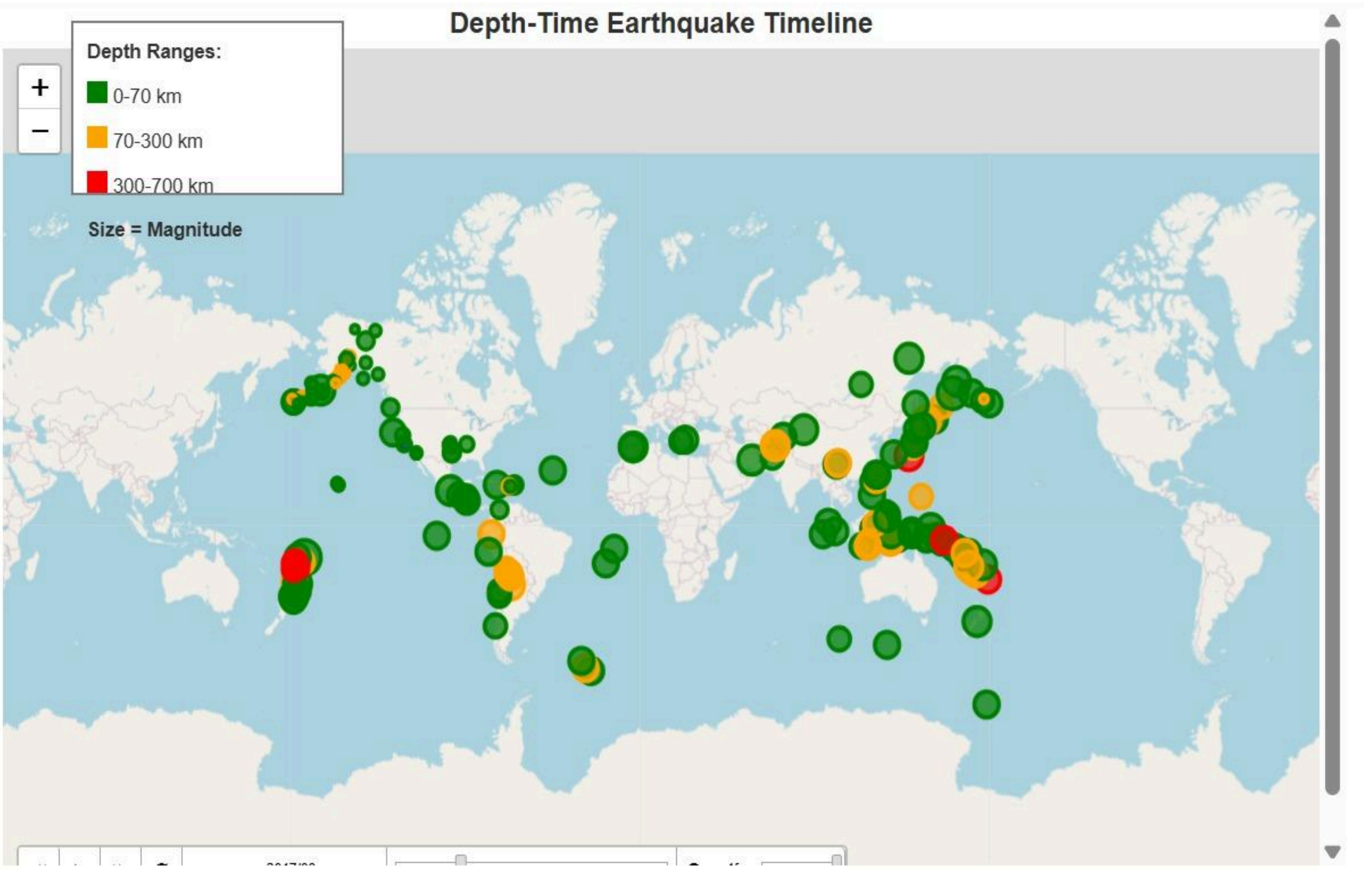
Subduction zones dominate as the highest-energy regions, mid-ocean ridges remain low-energy despite many small quakes, and stable continental interiors appear mostly dark. The second map shows seismic energy flux, or the average energy released per earthquake. High flux occurs where fewer but powerful earthquakes happen-such as Japan, Chile, Indonesia, and Alaska. Low flux characterizes divergent boundaries like mid-ocean ridges.

Together, the maps reveal where Earth releases the most seismic energy and how efficiently regions discharge accumulated tectonic stress.

3.3 Depth Timeline

3.3.1. Figure

[25]:



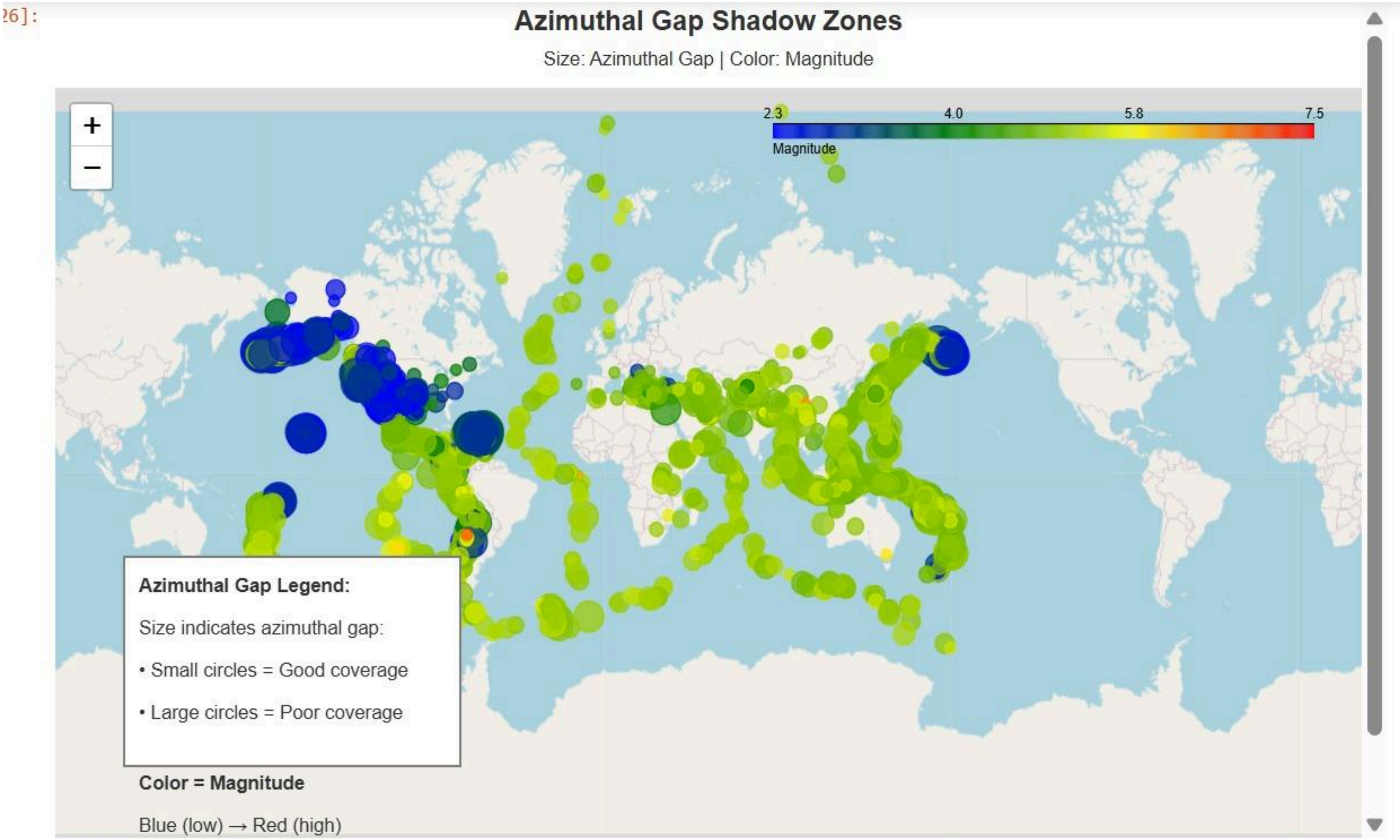
3.3.2. Description:

The Depth-Time Earthquake Timeline is an animated global map that shows how earthquakes evolve through time while highlighting their depths and magnitudes. Each earthquake appears as a colored circle at its geographic location, with the color representing its depth range. green for shallow quakes (0 -70 km), orange for intermediate-depth events (70 -300 km), and red for deep earthquakes (300- 700 km). The size of each circle encodes magnitude.

This visualization reveals temporal clustering, seismic cycles, and behavioral differences between shallow crustal activity and deeper subduction-zone events.

3.4 Azimuthal Gap Shadow Zones

3.4.1. Figure



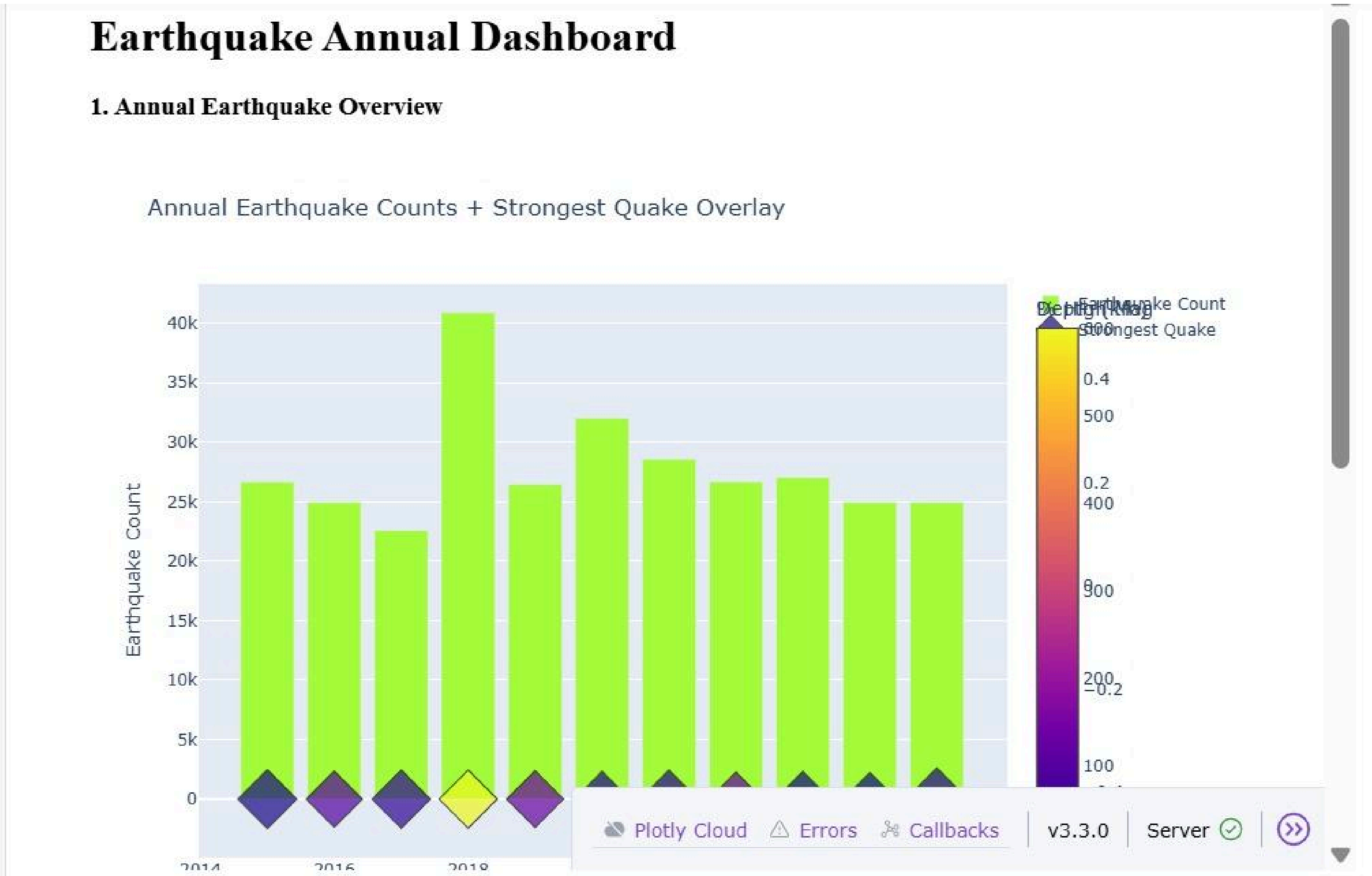
3.4.2. Description:

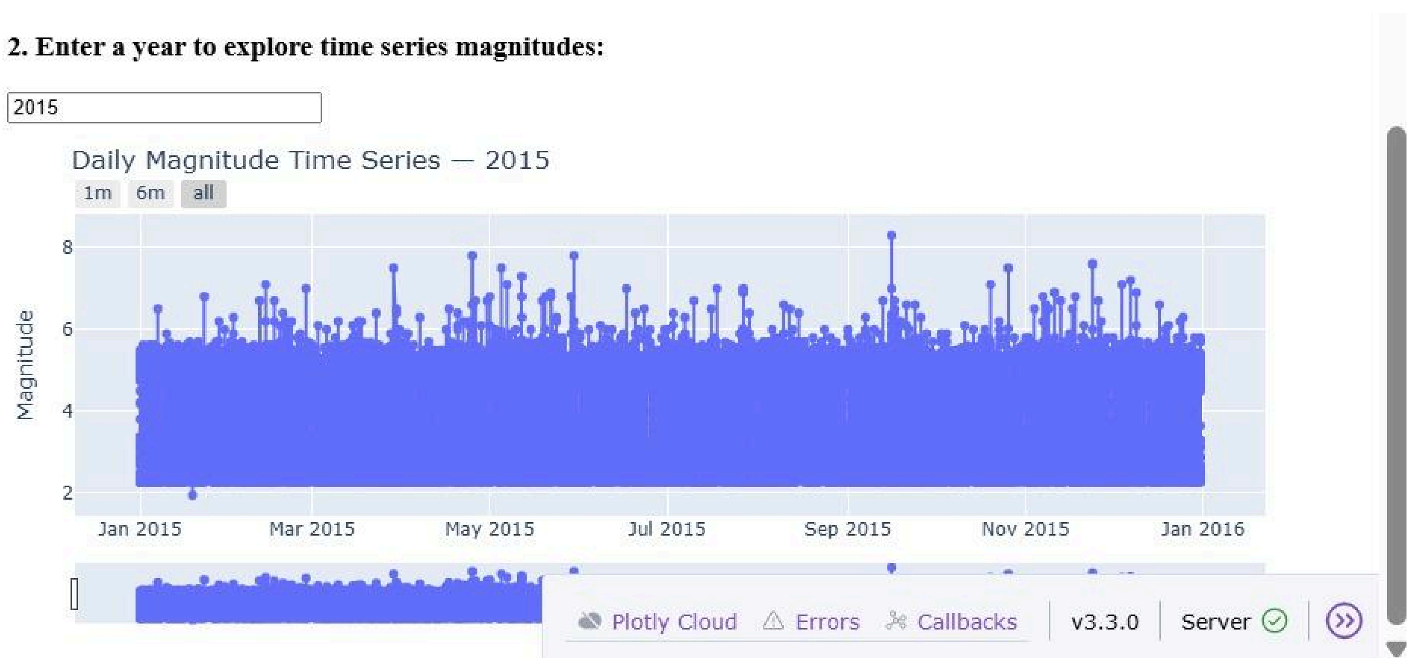
This visualization creates an interactive map displaying earthquake data with marker sizes proportional to azimuthal gap values larger circles indicate poorer seismic station coverage and greater location uncertainty while colors represent earthquake magnitudes. The map enables identification of regions with unreliable earthquake locations due to sparse monitoring networks while also conveying seismic intensity.

Users can zoom, pan, and click markers to view coordinates, magnitude, and azimuthal gap measurements, making it useful for assessing global seismic coverage quality.

3.5 Earthquake Annual

3.5.1. Figure





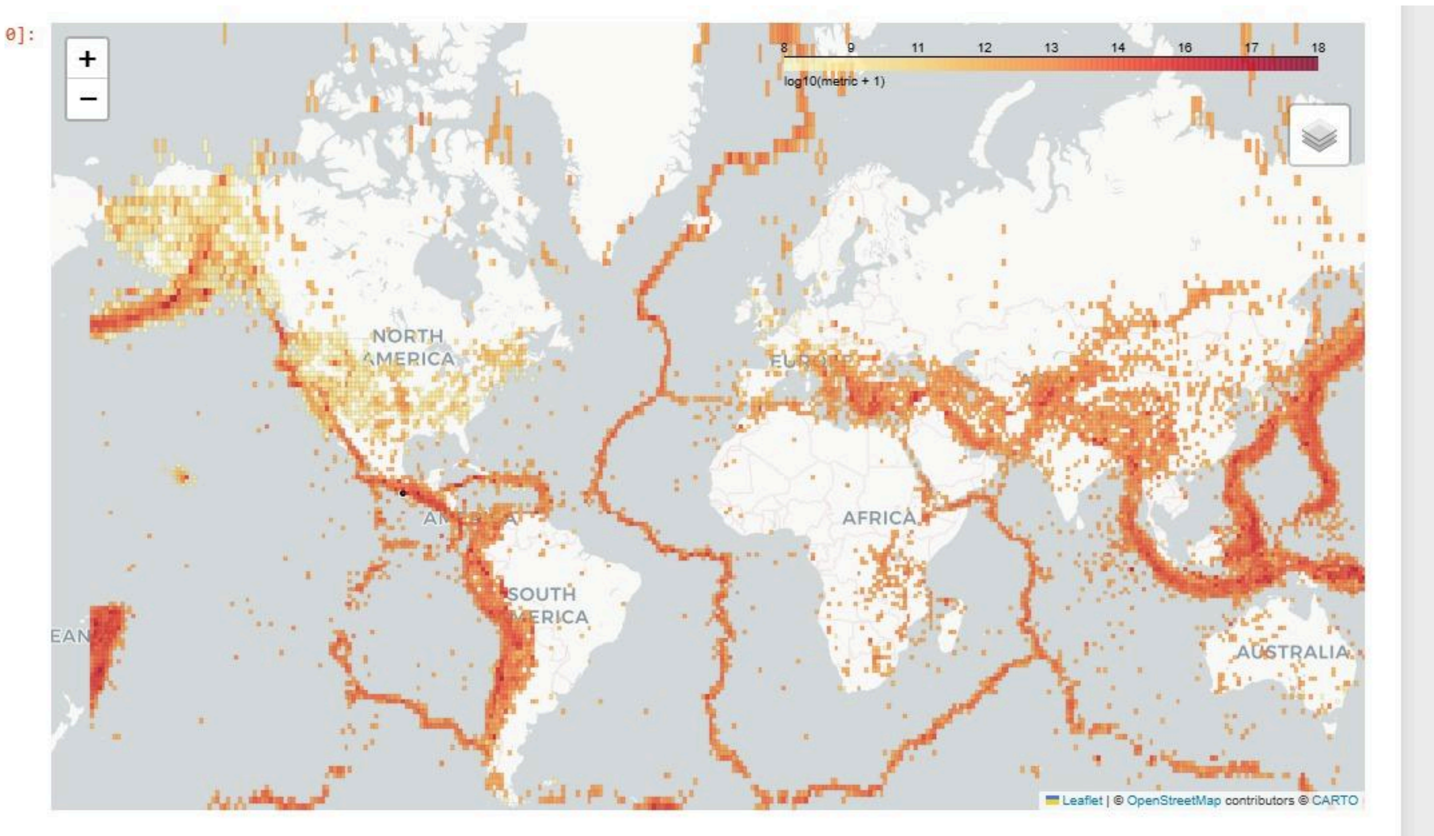
3.5.2. Description:

The dashboard reveals clear year-to-year variation in both the frequency and severity of earthquakes. Annual bars show how many events occurred and how the share of high-magnitude earthquakes (≥6.0) changes over time. Diamond markers highlight each year's strongest event and its depth. The time-series view allows closer inspection of any selected year, making it easy to spot periods of heightened activity or isolated major events.

Together, the visuals provide a coherent overview of long-term seismic trends while enabling detailed examination of short-term patterns.

3.6 Earthquake Grid

3.6.1. Figure

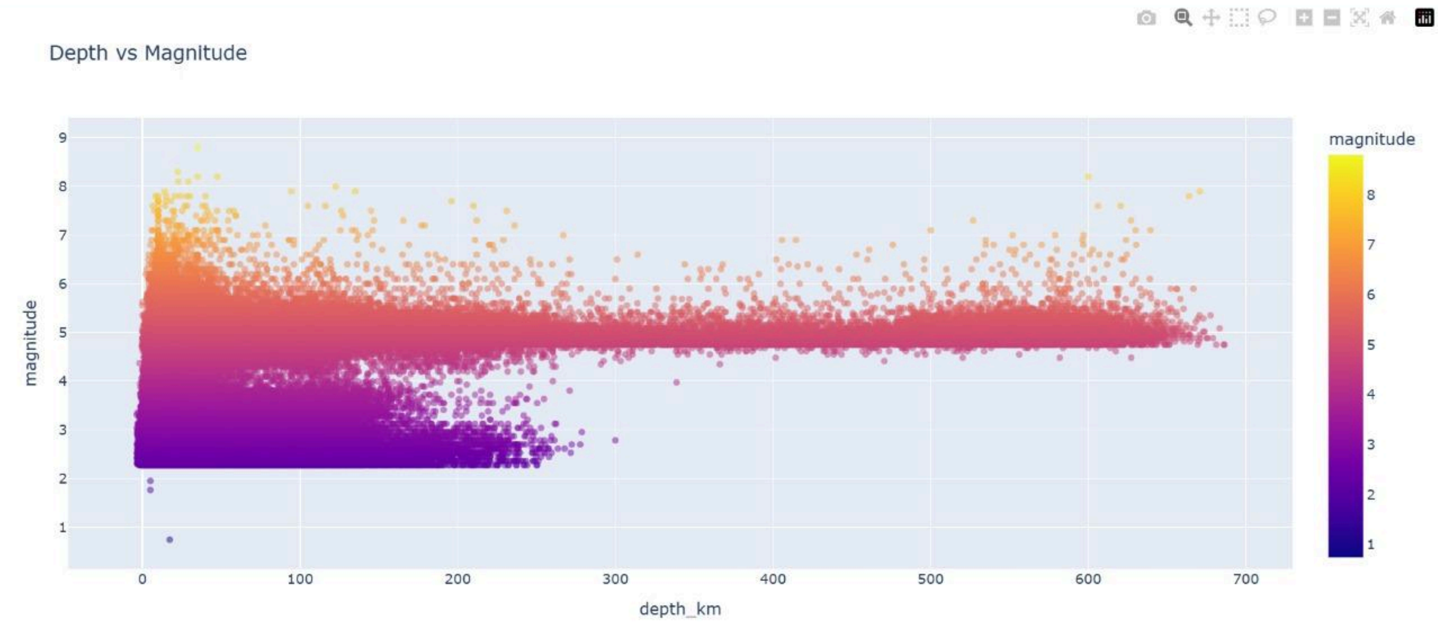


3.6.2. Description:

This visualization shows a global earthquake grid map, where the world is divided into equal sized geographic cells and each cell is color coded according to the number of recorded earthquakes. Darker red shading indicates regions with frequent seismic activity, while lighter tones represent fewer events. When the viewer hovers over a specific cell, an information box appears that displays the exact earthquake count for that location, which allows accurate and focused comparison between different areas. The map clearly emphasizes earthquake concentrations along active tectonic plate boundaries, including the Pacific Ring of Fire, parts of South Asia, and the western coasts of North and South America. In contrast, large portions of Africa, northern Europe, and central North America remain lightly shaded, reflecting relatively stable crustal environments. Overall, this visualization effectively communicates both the distribution and intensity of global seismic activity.

3.7 Depth vs Magnitude Analysis Using Scatterplot

3.7.1. Figure



3.7.2. Description:

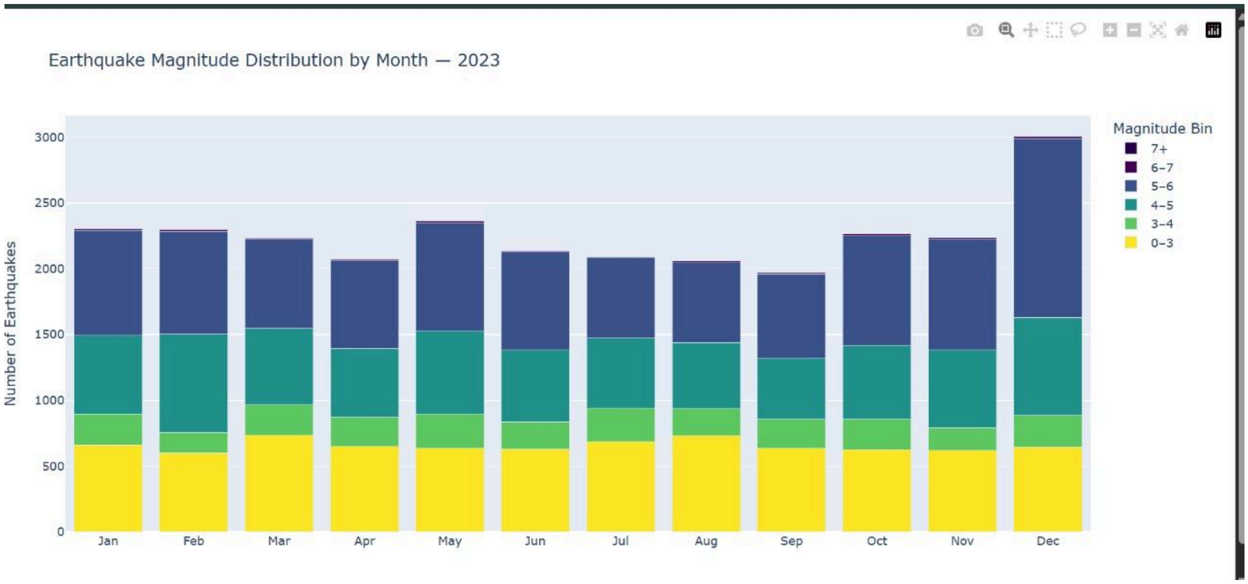
The scatter plot shows the relationship between earthquake depth (km) and magnitude for global seismic events. The following key patterns are visible:

- Most earthquakes occur at shallow depths (0–70 km)
- High-magnitude earthquakes (M ≥ 6) occur mostly at shallow to intermediate depths
- Deep earthquakes (300–700 km) are mostly moderate in magnitude
- The energy of earthquakes decreases as depth increases

This proves that the strongest and most destructive earthquakes happen at shallow to intermediate depths, making them the highest-risk seismic events.

3.8 Monthly Magnitude Distribution

3.8.1. Figure

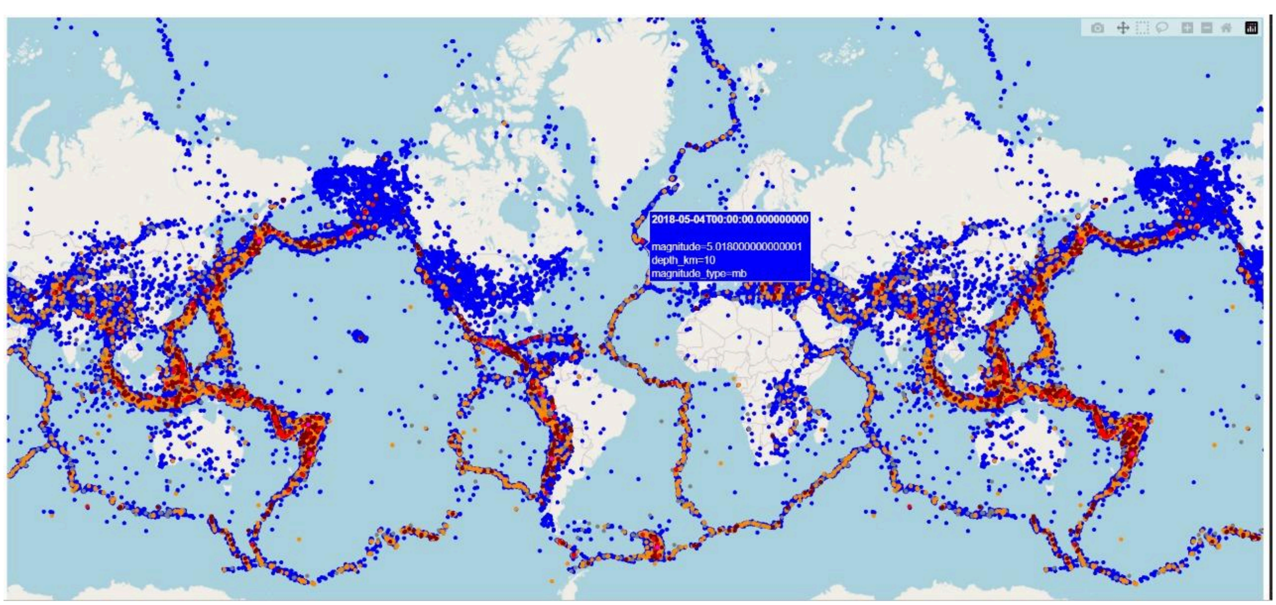


3.8.2. Description:

This stacked bar chart illustrates earthquake frequency throughout 2023 across different magnitude categories. Low-magnitude earthquakes consistently make up the largest share each month. Although monthly totals fluctuate, the distribution pattern remains stable. A notable spike occurs in December due to increases in the 4–5 and 5–6 magnitude ranges, indicating both higher seismic activity and more moderate earthquakes. High-magnitude events remain rare throughout the year.

3.9 How Often Earthquakes Occur

3.9.1. Figure



3.9.2. Description:

This world map shows where earthquakes happen and how frequently they occur. Some areas experience repeated seismic activity - such as Japan, Indonesia, and the west coast of South America—while regions like Africa, northern Europe, and central North America show minimal earthquake presence due to distance from tectonic boundaries. Subduction zones result in larger, deeper earthquakes, while mid-ocean ridges primarily produce frequent, shallow quakes.

4. Conclusion

Through detailed analysis and visualization, global earthquake behavior reveals strong correlations between tectonic settings, seismic depth, magnitude intensity, and geographic concentration. Most earthquakes occur along convergent and transform boundaries, and high-magnitude seismic events are predominantly shallow or intermediate in depth, making them more destructive. Temporal dashboards, spatial grids, completeness analyses, and magnitude distribution charts collectively enhance understanding of seismic risks and monitoring limitations.

These insights highlight the importance of reliable global seismic networks, energy-based assessments, and continuous visualization-driven research for improved resilience and hazard preparedness.

5. References

USGS Earthquake Catalog — https://earthquake.usgs.gov
Dataset Accessed: 2015–2025